

05 — Mission Crane 500 kW (cascade continues below)

Proves · The Mission row: a crane's full rating is its swing, and the cascade continues below it.

Mechanics this scenario turns on

- A **Mission** row takes H = the heavy consumer's full rating — a crane can start at any moment, so its whole draw is a potential peak.
- Cascade per row: H = what it wants · $I = \min(\text{remaining budget}, H)$ · $J = I \times \text{CoverageFactor}$ (covered) · $L = H - J$ (left to the gensets). Priority: DpReserve → DpDemand → Mission → Propulsion → Hotel. Invariant $\Sigma J + \Sigma L = \Sigma H$ — the sea sets the swing, the battery moves the split.
- Only the **uncovered** part rejoins the demand: $\text{propulsion}' = \text{propulsion} + L$, $\text{hotel}' = \text{hotel} + L$. Covered power (J) is never subtracted from anything.

Panels described below · Battery Contribution · Power Demands — the crane lands on the HOTEL side · Baseline & IL1 · Battery Benefit

Anything not described here behaves exactly as in `01-excel-baseline` — same plant, same hours; only what this scenario changes is worked through below.

Trust · verified against the reference workbook. These figures are proof.

Read after · scenario 04.

Test 01's vessel + **Mission Heavy-Consumer Max = 500** (battery 1260, Transit).

Battery Contribution (budget 1260)

The crane's H is its FULL kW — it can start at any moment, so the whole 500 is swing (Excel G7 = I3), covered at 50 %:

Row	H	I	J	L	Budget after
Mission (D=0.5)	500	500	250	250	760
Propulsion (D=0.35)	573.15	573.15	200.60	372.55	186.85
Hotel (D=0.05)	76	76	3.8	72.2	110.85 left

Tiles: **PS = 454.4** · **SR = 694.7**. Everyone got paid — the small crane doesn't starve the queue.

Power Demands — the crane lands on the HOTEL side

The crane is an electrical consumer on the switchboard, so its uncovered 250 goes to the hotel/aux side, not the shaft:

- Propulsion' = $11\,463 + 372.55 = \mathbf{11\,836}$ (identical to test 01!)
- Hotel' = $3\,800 + 250 + 72.2 = \mathbf{4\,122} \rightarrow \text{SG } 3\,250 + \text{AE } \mathbf{872}$
- Energy: ME **75 427 738** (same as 01) · AE $872.2 \times 5\,000 = \mathbf{4\,361\,000}$

Baseline & IL1

All FOCs shift up (AE side carries +250): 2.7108 / 2.7158 / **2.7181** (baseline, 3rd-from-worst) / 2.7193 / 2.7379. Baseline = **13 590.7 t/yr** (AE 872.2/12 000 = 7.3 %, AE fuel 1 005.2). IL1 = 2.7108×5 000 = **13 554.1** (AE 21.8 %, fuel 968.6). **Savings 36.6 t = \$28 542** — larger than 01 (21.2) because more aux-side load = more value in choosing the AE count. ME fuel is 12 585.6 in both — to the decimal the same as test 01: the ME never felt the crane.

Battery Benefit

Covered band ΣJ = 454.4 (vs 204.4 in 01) → benefit scales with it: **422.2 t/yr = \$329 352**. Crane + battery is a commercially strong pairing — the crane feeds the battery high-coverage material (50 % vs hotel's 5 %).

Takeaway: reserve routing is per-side (crane → hotel side), and battery value is proportional to the covered band.

KSailCalc manual test scenarios — calculation walkthrough · regenerated 2026-08-10 · numbers verified against commit 559aef2 and unchanged by the backend/client refactors (18 golden snapshots byte-identical)